# AI BASED EMERGENCY CALL AND SMS TRIAGE SYSTEM FOR DISASTER RESPONSE IN SRI LANKAN LOW RESOURCE LOCAL LANGUAGES

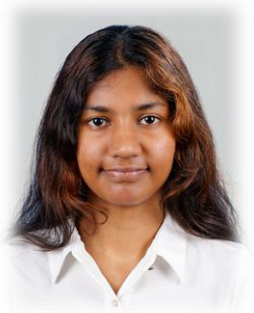
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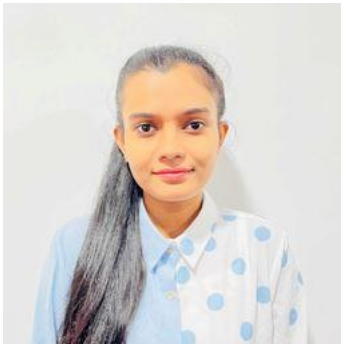
# Meet Our Team



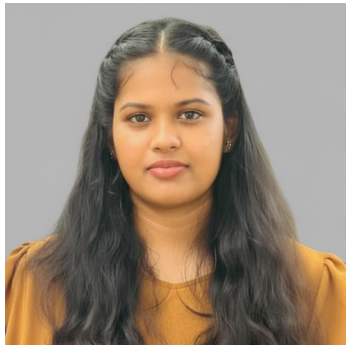
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# THE PROBLEM


Imagine rain pounding Sri Lanka for days.
invideo/it22129758
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by Getty Images

# IT22129758 | E.A.K. EDIRISINGHE

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Multilingual Emergency Conversational AI System For Rapid Clarification



# How the solution addresses the sub-problem /

## Sub-Problem

- Emergency callers may speak in Sinhala, Tamil, English, or mixed language.
- Callers may be panicked and unable to explain clearly.
- Important details may be missing during the call.
- Emergency responders may not get enough information quickly.




## Prototype / Solution Pipeline

1. Emergency Audio Call
2. Speech-to-Text
3. Emergency Information Extraction
4. Off-Topic Detection
5. Missing Field Detection
6. Clarification Question Generation
7. Question in Caller's Language

## How it helps

- Processes one emergency audio call step by step.
- Extracts incident type, location, people count, injury status, and trapped status.
- Detects missing important information.
- Generates a follow-up clarification question in the caller's language.

Multilingual Emergency Conversational AI System For Rapid Clarification

# User Requirements addressed & Design Contribution

## User Requirements Addressed

- Fast emergency understanding
- Multilingual support for Sinhala, Tamil, English, and mixed language
- Identifies missing emergency details
- Generates a suitable follow-up question
- Asks the question in the caller's language

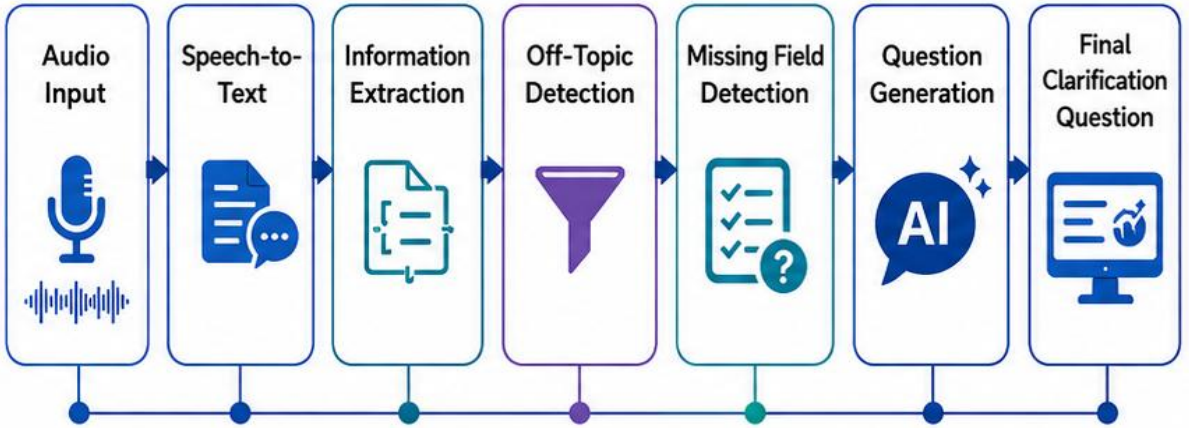






**Example:** If flood is reported but location is missing, the system asks for the location in the same language.

## Design Excellence / Contribution



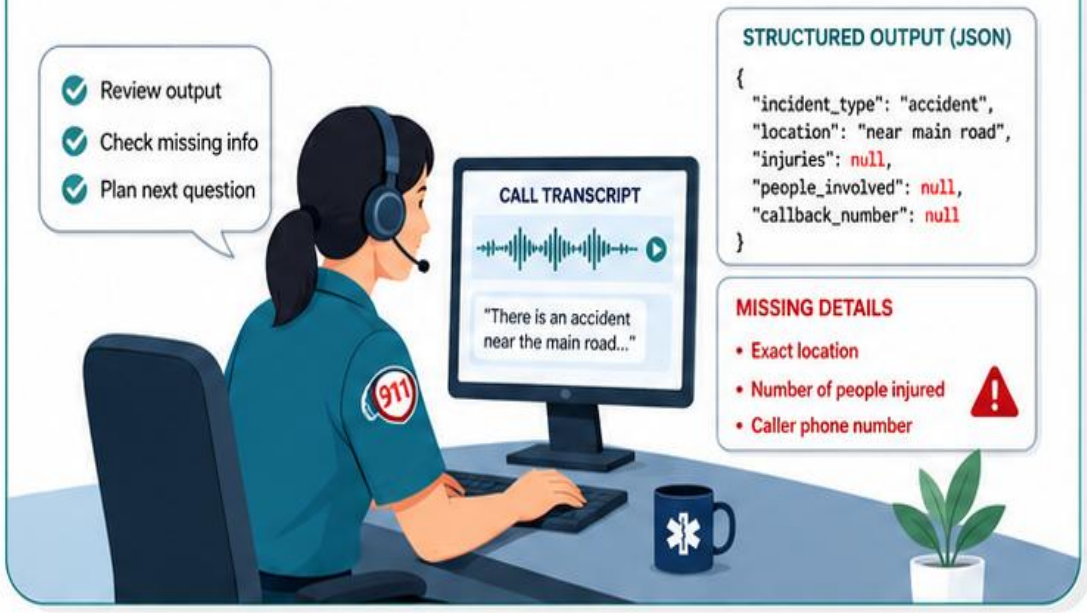

- Complete flow from audio input to final clarification question
- Separate modules for easier testing and improvement
- Backend pipeline is simple to explain and extend
- Supports future real-time Twilio call integration

Multilingual Emergency Conversational AI System For Rapid Clarification

# User Feedback & Evaluation (Prototype)

## Prototype Feedback

- Tested using sample emergency calls
- Helps identify missing critical information quickly
- Supports multilingual emergency understanding
- Reduces operator effort in deciding the next question




## Evaluation Summary

- Useful for fast emergency clarification
- Improves follow-up question selection
- Helps reduce manual decision-making
- Good prototype for future real-time deployment

### How It Works (Example Flow)

**Sample Call**
"There is an accident near the main road."

**Missing Details Detected**
Location, Injuries, Callback Number

**Follow-up Question Suggested**
"Can you please tell me the exact location of the incident?"

**Feedback (Prototype)**
Overall feedback is Positive

Multilingual Emergency Conversational AI System For Rapid Clarification

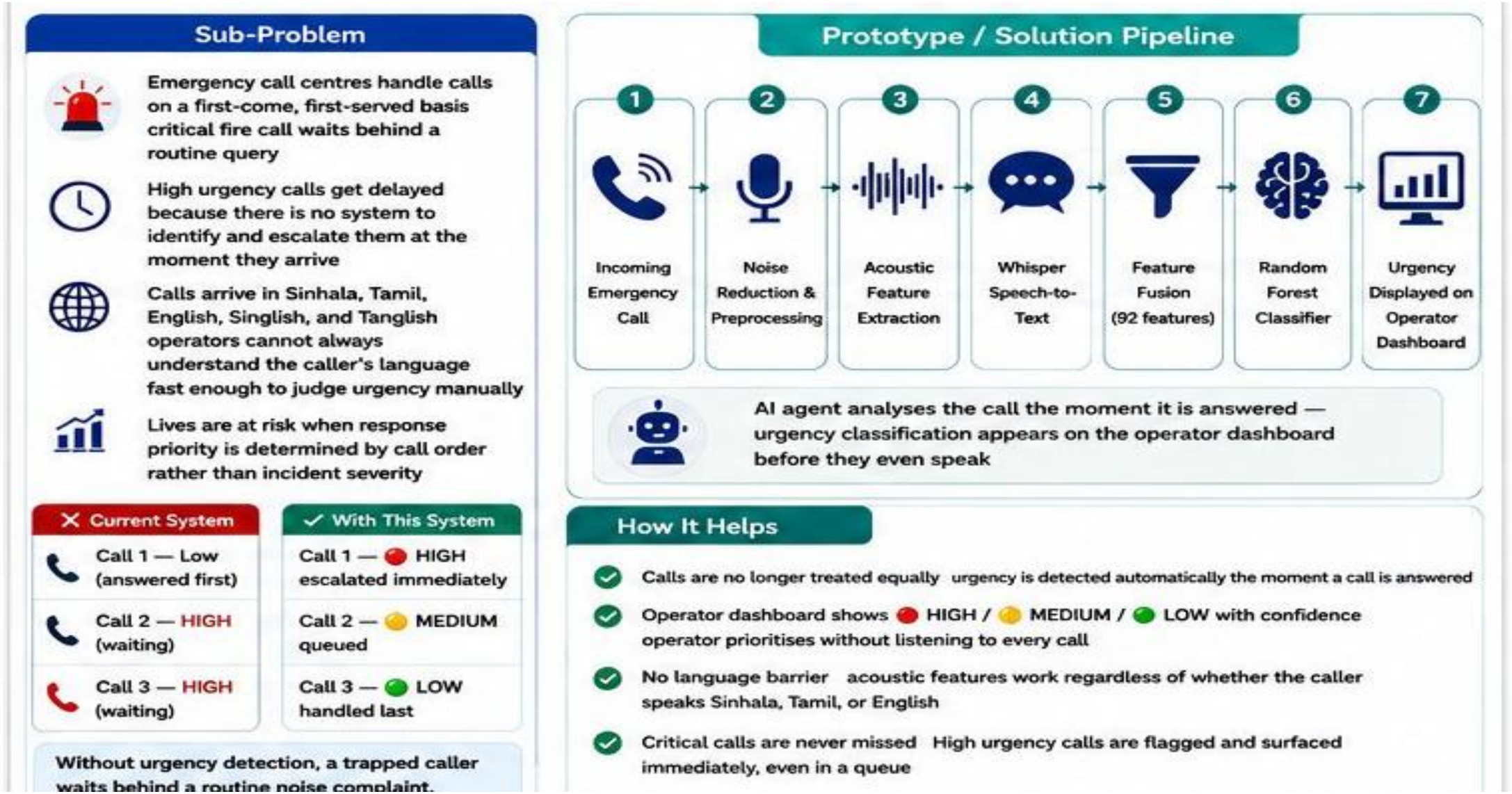
# Multimodal Urgency Detection Using Prosodic and Conversational Cues

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# Multimodal Urgency Classification Using Prosodic and Conversational Cues



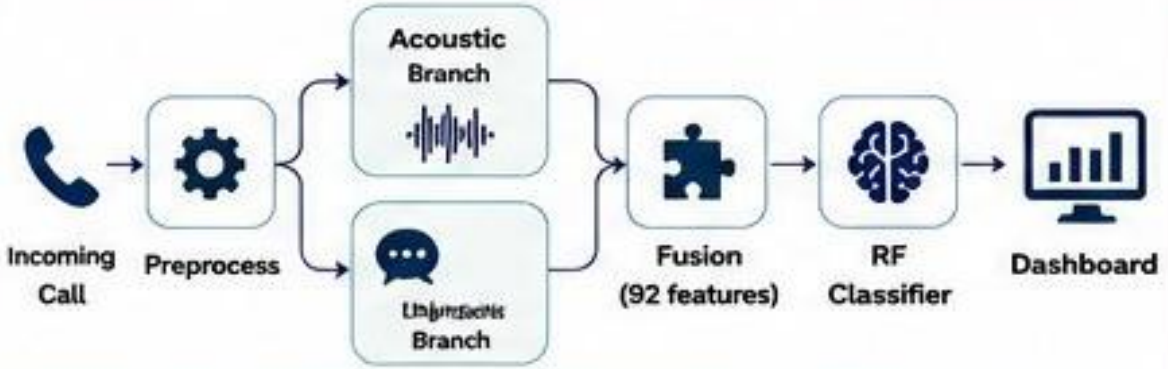
# Multimodal Urgency Classification Using Prosodic and Conversational Cues

## User Requirements Addressed

- **Priority-based call handling**
  system classifies urgency at call arrival so operators can attend to the most critical calls first, not the earliest
- **Multilingual**
  Sinhala, Tamil, English, Singlish, Tanglish handled through language-independent acoustic features
- **Works on panicked, unclear speech**
  prosodic features (pitch spike, voice tremor, energy burst) detect distress without needing clear words
- **Operator-readable output**
  confidence scores and colour coding on dashboard, no ML expertise required to interpret
- **Robust to noisy phone audio**
  noise reduction preproessing applied before classification

**Example:** Three calls arrive simultaneously. Caller A speaks calmly about a noise complaint. Caller B is a panicked Sinhala speaker trapped in a flood. Caller C reports a minor road issue. System instantly surfaces Caller B as 🔴 HIGH operator responds to them first.

## Design Excellence / Contribution




- **Real-time dual-branch classification**
  acoustic and lingusitic features extracted in parallel the moment a call is answered, not after it ends
- **Language-agnostic acoustic fallback**
  when Whisper fails on Sinhala (confirmed finding outputs hallucinated script), acoustic branch classifies independently no silent failures
- **SHAP explainability on dashboard**
  operator sees not just the urgency label but which vocal features triggered it auditable, trustworthy, deployable
- **Data-driven keyword discovery**
  TF-IDF analysis of real Whisper transcripts rebuilt the keyword list from actual evidence rather than manual guessing 85% of original keywords never appeared in transcripts

# Multimodal Urgency Classification Using Prosodic and Conversational Cues

## PROTOTYPE FEEDBACK

- ✓ **High urgency calls correctly flagged** — fire, flood, trapped caller scenarios classified in English and Tamil
- ✓ **Colour-coded output (🔴🟡🟢) understood immediately** without explanation
- ✓ **Whisper transcript display** confirmed what the system heard — increased tester confidence
- ✓ **Confidence percentage bars** helped operators understand certainty of prediction
- ✓ **No technical knowledge needed** — drag, drop, instant result

**DEMO RESULT — TAMIL HIGH URGENCY CALL**

*"நாங்கள் உள்ளே சிறிக்கிறோம்..."*

We are trapped inside...

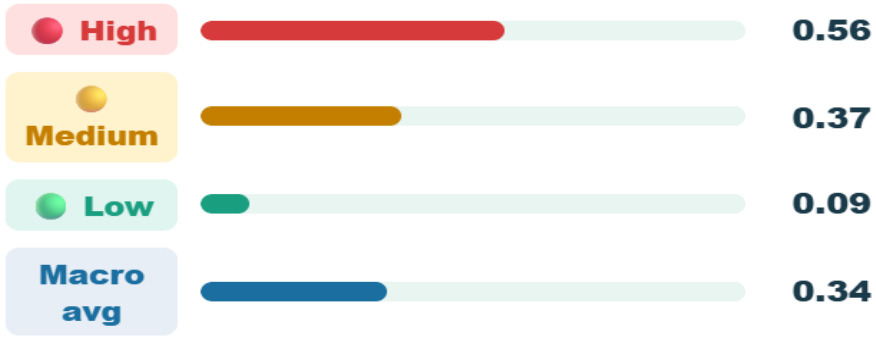
🔴 **HIGH** 81% confidence

Confidence

உதவி | உள்ளே | நாங்கள்

## EVALUATION SUMMARY + PP2 ROADMAP

### Model Performance (Random Forest — best overall)




RF outperformed SVM and Logistic Regression. Sinhala macro F1 = 0.461 on acoustics alone despite Whisper failure.

### PP2 Roadmap

- **Facebook MMS** for Sinhala ASR replacement
- **DMC real call data** — 500+ samples for retraining
- **Multilingual sentence embeddings** for linguistic branch
- **Speaker normalisation** to reduce demographic bias
- **Real-time streaming** classification pipeline

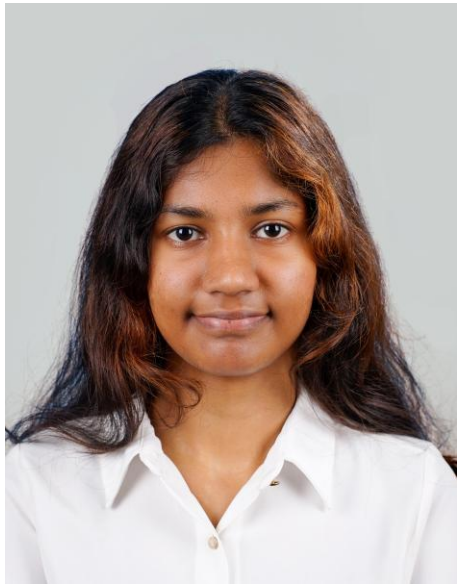
**Commercial Potential** ★★★★☆

Deployable as middleware in any emergency call centre — no hardware changes. Applicable across South Asian multilingual emergency services.

# Low-Bandwidth SMS Emergency Intelligence for Disaster Response

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# Low-Bandwidth SMS Emergency Intelligence for Disaster Response

## How the Solution Addresses the Sub-Problem / Prototype

| PROBLEM | EXACT SOLUTION |
| --- | --- |
| Disaster victims often send incomplete, emotional, and multilingual SMS messages during emergencies. | The system uses AI-powered NLP and multilingual processing to understand emotional and incomplete emergency messages in real time. |
| Existing emergency communication systems struggle to understand low-resource local languages such as **Sinhala**, **Tamil**, **Singlish**, and **Tanglish**. | The system supports **Sinhala**, **Tamil**, **Singlish**, **Tanglish**, and English language understanding using multilingual NLP and AI models. |
| Victims may not provide critical information such as location, number of people affected, or type of emergency. | The AI dynamically asks intelligent follow-up questions to collect missing emergency information automatically. |
| Traditional systems rely heavily on fixed keywords and forms. | The system uses hybrid AI reasoning and NLP instead of only fixed keyword-based processing. |
| Lack of intelligent systems capable of dynamically asking follow-up questions. | The multi-agent conversational AI generates context-aware follow-up questions based on missing information. |
| Difficulty in prioritizing emergency cases based on urgency and severity. | The priority agent classifies emergency situations into **LOW**, **MEDIUM**, and **HIGH** priority levels automatically. |
| Limited integration of multilingual NLP, AI reasoning, and real-time emergency triage. | The system combines multilingual NLP, generative AI, memory management, and emergency triage in a unified multi-agent architecture. |
| Emergency responders face challenges in rapidly identifying high-priority rescue cases. | The system automatically extracts and structures emergency information to help responders quickly **identify** critical rescue cases. |
| Existing systems provide limited conversational memory during emergency interactions. | The memory agent stores **previous** conversation data to support **accurate** multi-turn emergency communication. |
| Need for an AI-powered low-bandwidth emergency intelligence system for disaster response. | The proposed solution provides a **low-bandwidth AI-powered SMS and call** triage system for real-time disaster communication and rescue coordination. |

# Low-Bandwidth SMS Emergency Intelligence for Disaster Response

## USER REQUIREMENTS ADDRESSED

- People can send **emergency SMS or calls** in real time
- System supports **Sinhala, Tamil, Singlish, Tanglish** and English
- AI understands **emotional and incomplete** disaster messages
- Extracts **incident type, location, people affected** and needed help
- AI asks **follow-up questions** if information is missing
- System is **simple and user-friendly** for victims and rescue teams

## DESIGN EXCELLENCE / CONTRIBUTION

1. **Automatic Priority Classification**
   System automatically classifies emergency priority levels — enabling faster operator response without manual assessment

2. **Structured Disaster Information**
   AI organises and structures disaster information for faster rescue coordination — incident type, location, people count, injury status

3. **Low-Bandwidth Communication**
   Supports SMS and low-bandwidth channels for disaster situations with limited internet access — ensuring no victim is cut off

### SYSTEM OUTPUT — WHAT OPERATORS SEE

Incident Type | Location | People Count | Injury Status | Priority Level

# Low-Bandwidth SMS Emergency Intelligence for Disaster Response

## USER FEEDBACK & EVALUATION

### Residents / General Public

- ✓ Easy communication through **low-bandwidth SMS** and emergency calls
- ✓ AI gives **helpful and relevant responses** during disaster situations
- ✓ Supports **Sinhala and Tamil languages** — accessible to all communities

### Authorities / Disaster Management Teams

- ✓ **Better identification** of high-priority emergency cases
- ✓ Rescue teams can **identify urgent cases quickly** from the dashboard
- ✓ Emergency information becomes **organised automatically**
- ✓ Helps teams **understand disaster situations in real time**

★★★★☆ **Feedback: Positive**
Evaluated with residents, rescue teams & operators

## PROTOTYPE IMPROVEMENTS — PP2 ROADMAP

1. **Improve Prediction Accuracy**
   Enhance model training with larger multilingual emergency datasets to reduce classification errors under noisy or incomplete inputs

2. **Better Local Language Understanding**
   Expand coverage of regional language variations — Sinhala dialects, Tanglish, Singlish — with fine-tuned language models

3. **Improved AI Conversations**
   Refine follow-up question generation to be more contextually aware and natural — reducing caller confusion during emergencies

### EVALUATION SUMMARY

- Multilingual emergency messages handled correctly
- Follow-up questions generated in caller's language
- Priority triage displayed on operator dashboard
- Low-bandwidth SMS pathway confirmed working

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Multi-Source Urgency Based Incident Fusion ,UIRs & Real - Time Operator Dashboard

# Multi-Source Urgency Based Incident Fusion ,UIRs & Real - TimeOperator Dashboard

## How the solution addresses the sub-problem / prototype

Multi-Source Urgency Based Incident Fusion, UIRs & Real-Time Operator Dashboard

### Sub-problem

During a disaster, the same flood or fire generates hundreds of duplicate reports across voice calls and SMS

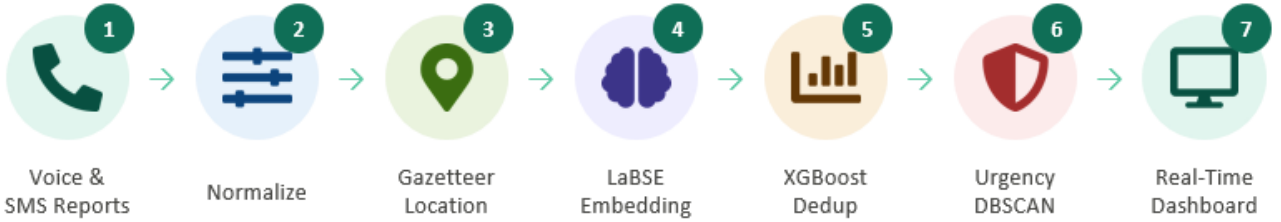
Reports arrive in Sinhala Unicode, Romanized Sinhala, and Tamil — the same incident in different scripts

Operators cannot tell which reports are genuine new incidents versus duplicates at the same location

A wrongly merged CRITICAL incident could send responders to the wrong location — potentially fatal

**Disaster Management Centre (DMC)**
Operators triaging duplicate reports with no tool support during active disasters

### Prototype / Solution Pipeline




### How it helps

- Fuses all duplicate reports about the same event into one clean Unified Incident Record (UIR)
- Resolves Sinhala and Tamil location names to GPS coordinates via multi-stage Gazetteer lookup
- CRITICAL incidents protected by a strict 5-condition gate — wrong merges are architecturally blocked
- Operators see one map pin per real incident; can Confirm, Split, Merge, or Dismiss each UIR

### User feedback

DMC IT team: collapsing duplicates into one record made incident scale immediately clear. Split, Merge, Confirm and Dismiss actions matched how operators work today.

### Suggestion

Add an audio alert when a new CRITICAL incident arrives so operators do not miss it while focused elsewhere on screen.

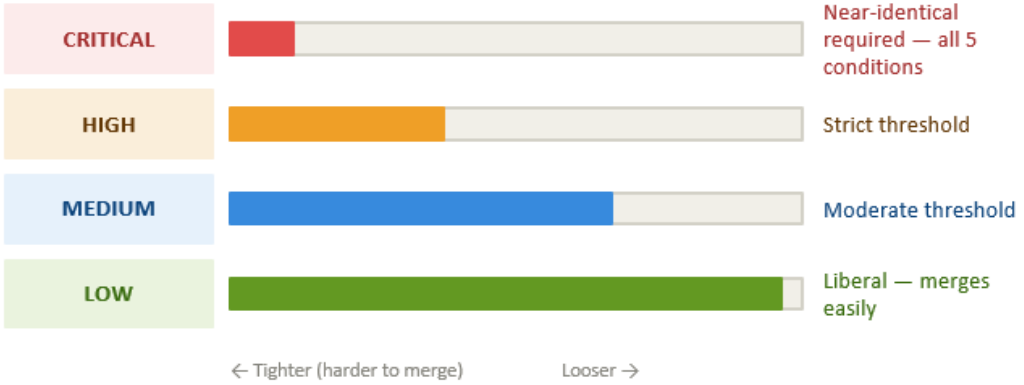
# Multi-Source Urgency Based Incident Fusion ,UIRs & Real - TimeOperator Dashboard

## Design excellence / Contribution

Multi-Source Urgency Based Incident Fusion, UIRs & Real-Time Operator Dashboard

**The gap:** All existing emergency clustering systems apply a **single fixed similarity threshold regardless of urgency.** The cost asymmetry — a false merge on a Critical incident (potentially fatal) vs a false split (one extra click) — is **never formalized as an architectural constraint. C4 fixes this.**

### Urgency-sensitive DBSCAN




**5-condition safety gate — CRITICAL only, all must pass**

- Semantic similarity above threshold
- Geographic proximity within tight radius
- Time gap ≤ 15 minutes
- At least 2 corroborating sources
- Incident type must match exactly

### XGBoost — why ML over fixed weights

Fixed weights (0.5 × semantic + 0.3 × geographic) cannot capture interaction effects. High semantic similarity combined with a location mismatch is a much weaker merge signal than either alone — fixed weights treat them independently. XGBoost learns these interaction features directly from labeled report pairs.

**Learned interaction features:**

- semantic × geographic
- semantic × temporal
- cross-lingual flag
- urgency difference
- address string similarity

### Key results

| 99.15% | < 2% | 97.3% |
|---|---|---|
| Precision | Critical FMR | Location coverage |

# Commercialization Plan

| 40%+ | 3–5× | 25 | 3+ |
|---|---|---|---|
| **Workload Reduction** | **Capacity Increase** | **Districts Covered** | **Languages Supported** |
| Automated triage, deduplication & prioritization | Simultaneous incidents managed without extra staffing | National DMC deployment across all Sri Lankan districts | Sinhala · Tamil · English · Code-mixed dialects |

## 6.1 TARGET MARKET & CUSTOMER PERSONA

### Disaster Management Centre (DMC)

Primary Customer · Sri Lanka National Body

Coordinates disaster preparedness & response across all 25 districts. Operationally confirmed need for this system.

### Secondary Target Customers

- Ministry of Public Security
- Police Emergency 119
- Provincial Disaster Management Units
- Colombo Municipal Council Emergency Services

### Longer-term International Market

Nepal · Bangladesh · Myanmar · SAARC Region

UNDP · Red Cross

## 6.2 VALUE PROPOSITION

- **First and only** AI emergency triage system supporting Sinhala, Tamil, and code-mixed Sri Lankan communication
- Reduces operator workload by **40%+** by automating report triage, deduplication & prioritization
- Reduces **critical missed-response risk** through false-negative-optimized urgency detection
- Enables operators to manage **3–5×** more simultaneous incident reports **without additional staffing**
- Works on **low bandwidth and SMS** — critical for disaster contexts where voice networks may be saturated